

A Mechanical Extension of General Relativity: The Singularity in Equilibrium

Robin Skavberg

Independent Researcher

ORCID: [0009-0003-9126-6196](https://orcid.org/0009-0003-9126-6196)

Email: skavberg@gmail.com

February 2, 2026

Abstract

General Relativity (GR) meets a formal limit at the singularity, where density and curvature diverge toward infinity. This paper proposes a structural extension to the Einstein-Hilbert framework by applying the Stiffness Matching Principle (SMP), where the gravitational constant is a dynamic ratio $G(P_m) = c^2/(8\pi P_m)$. We demonstrate that as field pressure P_m approaches infinity, the coupling vanishes ($G \rightarrow 0$), causing the spacetime manifold to reach a state of mechanical saturation. This stabilizes the singularity into a physical state of equilibrium with a finite volume. We further show that Hawking radiation emerges as a mechanical pressure-equalization process and that this framework resolves the Black Hole Information Paradox by ensuring quantum states are preserved within a saturated geometric core.

1 Introduction: The Singularity in Equilibrium

The mathematical prediction of a singularity in standard General Relativity suggests a breakdown of causality. Building upon the mechanical derivation of G [1], we propose that the singularity exists as a stable state of equilibrium sustained by mechanical saturation. Gravity is viewed as the mechanical manifestation of the manifold's resistance to compression; the singularity represents the terminal threshold where matter-field pressure matches the intrinsic vacuum stiffness (8π).

While early variable- G theories (Brans-Dicke [6], Dirac [10]) focused on cosmological time-scales, and regular black hole models (Bardeen [8], Hayward [9]) provided geometric workarounds for singularities, the SMP [1] provides a unifying mechanical explanation based on Sakharov's conception of vacuum elasticity [7].

2 Formalizing the Dynamic Coupling

The SMP serves as the analytical tool for observing the singularity in equilibrium, allowing G to respond to the tensor strength of the field. Using the multiplicative ratio defined in [1]:

$$G(P_m) = \frac{c^2}{8\pi P_m} \tag{1}$$

As $P_m \rightarrow \infty$, the coupling constant respects the vanishing limit:

$$\lim_{P_m \rightarrow \infty} G(P_m) = 0 \tag{2}$$

This vanishing of G implies that at the core of a singularity, the manifold becomes infinitely "stiff," effectively halting further gravitational collapse.

3 Modified Einstein Field Equations

By substituting the dynamic coupling into the Einstein Field Equations (EFE) [5], we define the effective coupling κ_{eff} :

$$\kappa_{eff}(P_m) = \frac{8\pi G(P_m)}{c^4} = \frac{1}{c^2 P_m} \quad (3)$$

The modified EFE is expressed as:

$$G_{\mu\nu} = \left(\frac{1}{c^2 P_m} \right) T_{\mu\nu} \quad (4)$$

In the extreme high-pressure regime of a singularity, $T_{\mu\nu}$ and P_m scale proportionally. Consequently, the curvature tensor $G_{\mu\nu}$ does not diverge but reaches a saturation constant:

$$\lim_{P_m \rightarrow \infty} G_{\mu\nu} = \Lambda_{sat} \quad (5)$$

This proves the existence of a **Singularity in Equilibrium**, a finite-density core where geometry and energy density are mechanically balanced.

4 Mechanical Proof of Hawking Radiation

Under the SMP, Hawking radiation [2] is the thermodynamic consequence of the pressure gradient between the saturated core (P_m) and the ambient vacuum stiffness. The Hawking temperature T_H is re-expressed by substituting the dynamic G :

$$T_H = \frac{\hbar c^3}{8\pi G(P_m) M k_B} = \frac{\hbar c P_m}{M k_B} \quad (6)$$

Radiation is thus the process of the singularity "bleeding" field pressure to maintain its equilibrium density. This ensures that information is encoded within the finite volume of the saturated core rather than being destroyed [3].

5 The Mechanical Floor: Planck Scale Limits

A primary consequence of $G(P_m) \rightarrow 0$ is the redefinition of the effective Planck length $\ell_{P(eff)}$:

$$\ell_{P(eff)} = \sqrt{\frac{\hbar G(P_m)}{c^3}} = \sqrt{\frac{\hbar}{8\pi c P_m}} \quad (7)$$

As P_m approaches saturation, $\ell_{P(eff)}$ provides a "mechanical floor." The singularity is thus bounded within a finite volume $V_{min} \approx (\ell_{P(sat)})^3$, satisfying Bekenstein-Hawking boundary conditions without mathematical infinity.

6 Observational Verification: S2 Orbit

The SMP predicts a measurable damping in the Schwarzschild precession of the star S2 [4]. Table 1 illustrates the predicted deviation from GR as the star approaches higher field-pressure regions near the galactic center.

$$\Delta\phi_{SMP} = \Delta\phi_{GR} \left(\frac{c^2}{8\pi G_0 P_m} \right) \quad (8)$$

Table 1: Comparative Precession: GR vs. SMP Models

r (Schwarzschild radii)	P_m (Relative)	$\Delta\phi_{GR}$	$\Delta\phi_{SMP}$	Deviation
1000	1.0	12.10''	12.09''	0.08%
500	2.0	24.20''	23.98''	0.91%
100	10.0	121.0''	114.2''	5.62%
50	50.0	242.0''	205.7''	15.0%

7 Conclusion

By establishing G as a dynamic variable that vanishes at the limit of infinite pressure, we resolve the instabilities of General Relativity. The singularity in equilibrium provides a stable, testable extension to GR that preserves the fundamental laws of both gravity and quantum mechanics.

AI Transparency Statement

The author utilized Gemini (Google) for assistance in LaTeX document structuring, bibliographical formatting, and mathematical validation of the dynamic limits. The core theoretical framework and the Stiffness Matching Principle (SMP) are the original work of the author.

References

- [1] Skavberg, R. (2026). *A Mechanical Route to G: Matching Dark-Matter Field Pressure to the Einstein-Hilbert Coupling*. Zenodo. DOI: 10.5281/zenodo.18393277.
- [2] Hawking, S. W. (1976). *Breakdown of predictability in gravitational collapse*. Physical Review D, 14(10), 2460. DOI: 10.1103/PhysRevD.14.2460.
- [3] Susskind, L. (1995). *The World as a Hologram*. Journal of Mathematical Physics, 36(11), 6377. DOI: 10.1063/1.531249.
- [4] GRAVITY Collaboration. (2020). *Detection of the Schwarzschild precession in the orbit of the star S2*. Astronomy & Astrophysics, 636, L5. DOI: 10.1051/0004-6361/202037813.
- [5] Einstein, A. (1915). *Die Feldgleichungen der Gravitation*. Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften (Berlin), 844–847. Bibcode: 1915SPAW.....844E.
- [6] Brans, C., and Dicke, R. H. (1961). *Mach's Principle and a Relativistic Theory of Gravitation*. Physical Review, 124(3), 925. DOI: 10.1103/PhysRev.124.925.

- [7] Sakharov, A. D. (1967). *Vacuum Quantum Fluctuations in Curved Space and the Theory of Gravitation*. Soviet Physics Doklady, 12, 1040. Bibcode: 1968SovPh..12.1040S.
- [8] Bardeen, J. M. (1968). *Non-singular general relativistic gravitational collapse*. In: Proceedings of International Conference GR5, Tbilisi, Georgia, 174. Bibcode: 1968gr5..conf..174B.
- [9] Hayward, S. A. (2006). *Formation and evaporation of regular black holes*. Physical Review Letters, 96(3), 031103. DOI: 10.1103/PhysRevLett.96.031103.
- [10] Dirac, P. A. M. (1937). *The Cosmological Constants*. Nature, 139, 323. DOI: 10.1038/139323a0.